

CV/Resume

KRISH DESAI, PHD

Employment: Data Scientist

Bank of America

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One Bryant Park,

New York, NY 10036,

United States.

E-mail: krish@desai.ml

Website: https://www.desai.ml

Profiles:     

Education

August 2020 – August 2025 University of California, Berkeley

Ph.D., Physics

August 2017 – May 2020 Yale University

M.S., Mathematics

B.S., Mathematics (Intensive)

B.S., Physics (Intensive)

Professional Experience

May 2026 – Present Bank of America

Data Scientist, Data and AI Group

October 2020 – May 2026 Lawrence Berkeley National Lab

Machine Learning Researcher

May 2025 – December 2025 Emissary AI

Machine Learning Engineer

August 2020 – December

2024

Associate Instructor

June 2023 – August 2023 Bridgewater Associates

Investment Analyst Intern

May 2022 – August 2022 Microsoft Research

PhD Research Intern

May 2020 – August 2020 Purple Gaze Inc.

Software Developer Intern

Publications

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- September 2026 Machine Learning-based Unfolding for Cross Section Measurements in the Presence of Nuisance Parameters
- Huanbiao Zhu, **Krish Desai**, Mikael Kuusela, Vinicius Mikuni, Benjamin Nachman, and Larry Wasserman
- Annals of Applied Statistics*
- [PDF](#) [BIBLATEX](#)
- September 2026 Reweighting Adversarial Networks for Unbinned Unfolding
- Umar Sohail Qureshi, **Krish Desai**, Jesse Thaler, and Benjamin Nachman
- Physical Review X*
- [PDF](#) [BIBLATEX](#)
- June 2026 Reweighting Adversarial Networks for Unbinned Unfolding
- Umar Sohail Qureshi, **Krish Desai**, Benjamin Nachman, and Jesse Thaler
- Conference on Physics and AI (PAI26), Stanford University*
- [PDF](#) [BIBLATEX](#)
- March 2026 Neural Posterior Unfolding
- Fernando Torales Acosta, Jay Chan, **Krish Desai**, Vinicius Mikuni, Benjamin Nachman, Jingjing Pan, Francesco Rubbo

November 2025 Unsupervised Evaluation of Multi-Turn Objective-Driven Interactions

Emi Soroka, Tanmay Chopra, **Krish Desai**,
and Sanjay Lall

arXiv:2511.03047

October 2025 Unbinned Inference with Correlated Events

Krish Desai, Owen Long, and Benjamin Nachman

European Physical Journal C

August 2025 Machine Learning Methods for Cross Section Measurements

Krish Desai

University of California, Berkeley

PhD (Physics) Dissertation

December 2024 Multidimensional Deconvolution with Profiling

Huanbiao Zhu, **Krish Desai**, Mikael Kuusela, Vinicius Mikuni, Benjamin Nachman, and Larry Wasserman

*Thirty-Eighth Annual Conference on Neural Information
Processing Systems (NeurIPS), ML4PS Track*

[PDF](#) [BIBLATEX](#)

December 2024 Neural Posterior Unfolding

Fernando Torales Acosta, Jay Chan, **Krish Desai**,
Vinicius Mikuni, Benjamin Nachman, and Jingjing Pan

*Thirty-Eighth Annual Conference on Neural Information
Processing Systems (NeurIPS), ML4PS Track*

[PDF](#) [BIBLATEX](#)

December 2024 Moment Unfolding: Moment extraction using an unfolding
protocol without binning

Krish Desai, Benjamin Nachman, and Jesse Thaler
Physical Review D

[PDF](#) [BIBLATEX](#)

December 2022 Deconvolving Detector Effects for Distribution Moments

Krish Desai, Benjamin Nachman, and Jesse Thaler
*Thirty-Sixth Annual Conference on Neural Information
Processing Systems (NeurIPS), ML4PS Track*

[PDF](#) [BIBLATEX](#)

May 2022 SymmetryGAN: Symmetry Discovery with Deep Learning

Krish Desai, Benjamin Nachman, and Jesse Thaler

December 2021 Oblivious points on translation surfaces

Ian Adelstein, **Krish Desai**, Anthony Ji, and
Grace Zdeblick

Journal of Geometry

December 2021 Symmetry Discovery with Deep Learning

Krish Desai, Benjamin Nachman, and Jesse Thaler

*Thirty-Fifth Annual Conference on Neural Information
Processing Systems (NeurIPS), ML4PS Track*

April 2020 Padé Approximants and the Anharmonic Oscillator

Krish Desai

Yale University

BS and MS (Mathematics, Intensive) Thesis

Honors & Fellowships

2025 Full Member of Sigma Xi

Sigma Xi

Election to the nation's oldest academic honor society

2020 Howard L. Schultz Prize

Yale University

The most outstanding graduating senior in physics

2020 Distinction in Mathematics

Yale University

2020 Distinction in Physics

Yale University

2020 Lifetime Member of Sigma Pi Sigma

Sigma Pi Sigma

Election to the national physics honor society

2020 George J. Schulz Summer Fellowship

Yale University Chapter

2019 Howard Robert Topol Fellowship

Yale University

2018 James A. Helzer Fellowship

Yale University

2018 SUMRY Fellowship

Yale University

2018 Yale First-Year Research Fellowship

Yale University

2018 Benjamin F. Barge Prize

Yale University

Most outstanding first-year student in mathematics

Talks

December 2024 Neural Posterior Unfolding

*Thirty-Eighth Annual Conference on Neural Information
Processing Systems*

Vancouver, Canada

December 2024 Multidimensional Deconvolution with Profiling

*Thirty-Eighth Annual Conference on Neural Information
Processing Systems*

Vancouver, Canada

July 2024 CMS Seminar

CERN

Geneva, Switzerland

June 2024 Moment extraction using an unfolding protocol without
binning

PHY-STAT Unfolding Conference

Paris, France

December 2022 Deconvolving Detector Effects for Distribution Moments

*Thirty-Sixth Annual Conference on Neural Information
Processing Systems*

New Orleans, LA

November 2022 Moment Unfolding with Deep Learning

Machine Learning for Jets

New Brunswick, NJ

June 2022 High Energy Physics Seminar

Korea Institute for Advanced Study

Seoul, South Korea

April 2022 Moment Unfolding using Deep Learning

American Physical Society

New York, NY

December 2021 Symmetry Discovery with Deep Learning

*Thirty-Fifth Annual Conference on Neural Information
Processing Systems*

Virtual

November 2021 Introduction to Symmetry Discovery and Deep Learning

Berkeley Compass Lecture

Berkeley, CA

October 2021 Symmetry Discovery and Deep Learning

Clarifai Perceive Deep Learning AI Conference

Virtual

July 2021 SymmetryGAN

Machine Learning for Jets

Heidelberg, Germany

April 2021 Symmetry Discovery

*Lawrence Berkeley National Laboratory, Physics Division
Seminar*

Virtual

April 2021 Symmetry Discovery using Machine Learning

American Physical Society

Virtual

August 2019 Oblivious Points on Translation Surfaces

Young Mathematicians Conference

Columbus, OH

August 2019 Closed Geodesics on Translation Surfaces

Massachusetts Undergraduate Research Conference

Amherst, MA

Teaching

Fall 2024 PHYSICS 88: Data Science Applications in Physics

Associate Instructor

University of California, Berkeley

Fall 2024 PHYSICS 77: Introduction to Computational Techniques in
Physics

Associate Instructor

University of California, Berkeley

Summer 2024 PHYSICS 77: Introduction to Computational Techniques in
Physics

Associate Instructor

University of California, Berkeley

Fall 2023 PHYSICS 89: Introduction to Mathematical Physics

Associate Instructor

University of California, Berkeley

Summer 2023 PHYSICS 77: Introduction to Computational Techniques in
Physics

Associate Instructor

University of California, Berkeley

Fall 2022 PHYSICS 88: Data Science Applications in Physics

Associate Instructor

University of California, Berkeley

Fall 2022 PHYSICS 77: Introduction to Computational Techniques in
Physics

Associate Instructor

University of California, Berkeley

Spring 2022 PHYSICS 8B: Introductory Physics

Head Associate Instructor

University of California, Berkeley

Summer 2021 PHYSICS 7B: Physics for Scientists and Engineers

Associate Instructor

University of California, Berkeley

Spring 2021 PHYSICS 7B: Physics for Scientists and Engineers

Associate Instructor

University of California, Berkeley

Fall 2020 PHYSICS 7B: Physics for Scientists and Engineers

Associate Instructor

University of California, Berkeley

Leadership & Service

2026	Invited Peer Reviewer	Archives of Computational Methods in Engineering
2025, 2026	Invited Peer Reviewer	Nature, Scientific Reports
2025	Invited Peer Reviewer	Journal of High Energy Physics
2023-2025	Graduate Mentor	Math and Physical Sciences Scholars, UC Berkeley
2022, 2024, 2025, 2026	Invited Peer Reviewer	NeurIPS
2022-2024	Campus Affairs Vice President, and Rules Officer	UC Berkeley Graduate Assembly
2023	Research Mentor	UC Berkeley Physics Innovators Program
2021 – 2024	Committee Member	UC Berkeley Physics Faculty Search
2018-2019	Vice President	Yale Political Union

Skills

Programming Languages	Python · C · Julia · Bash
LLMs & Agentic Systems	Agentic Orchestration · LangGraph · Retrieval-Augmented Generation · LLM Evaluation · vLLM · Fine-Tuning · Reinforcement Learning (GRPO)
Machine Learning	PyTorch · JAX · Keras · scikit-learn · Deep Learning · Generative Models · Normalizing Flows · Adversarial Networks
Statistics & Data Science	Statistical Inference · Bayesian Inference · Hierarchical Models · Uncertainty Quantification · Monte Carlo · Unfolding · Deconvolution
Physics & Mathematics	Particle Physics · Quantum Field Theory · Stochastic Calculus · Operator Theory · Differential Geometry
Computing	HPC · Distributed Multi-GPU Training · Git · Docker
Languages	English · French · Hindi · Gujarati